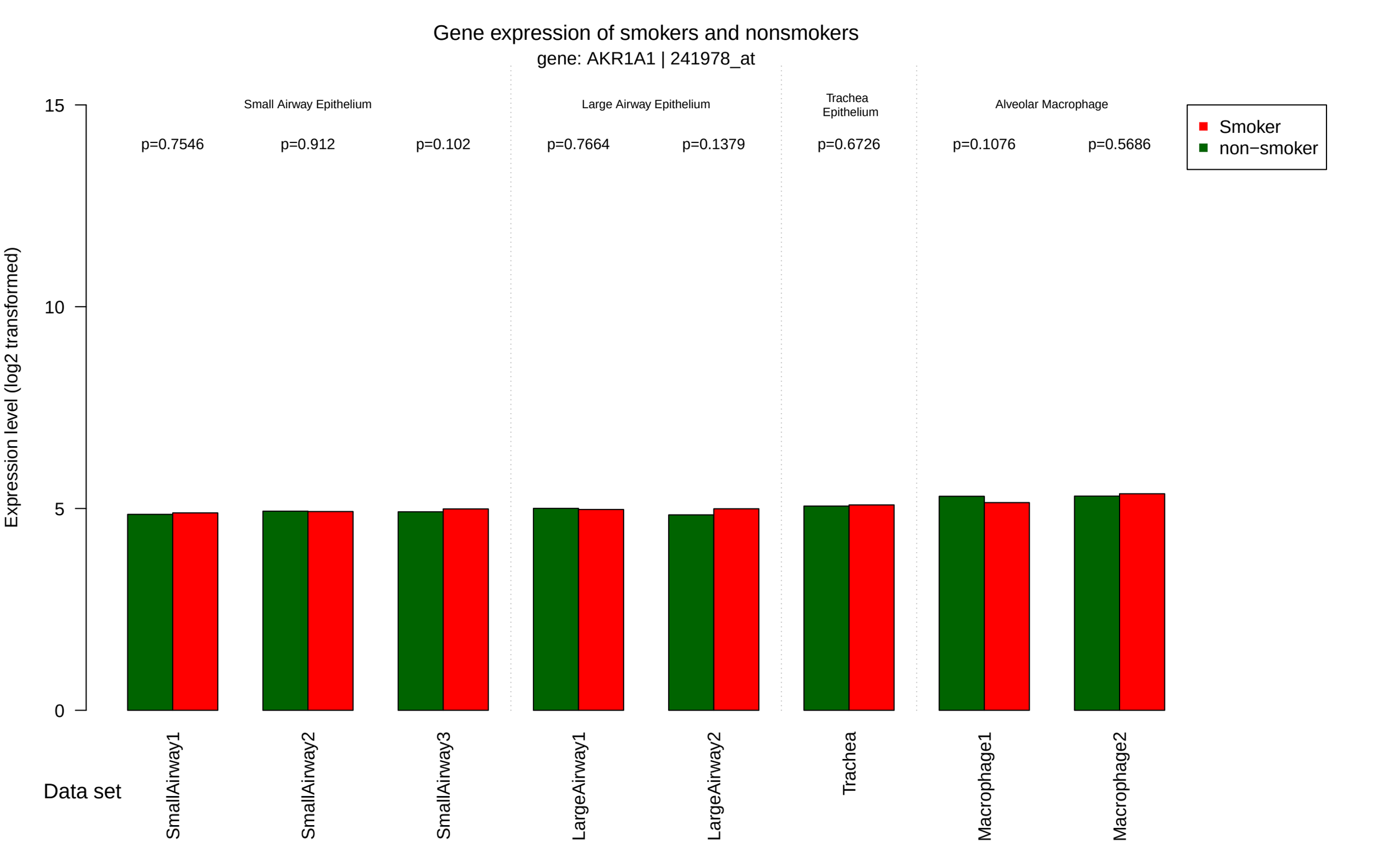
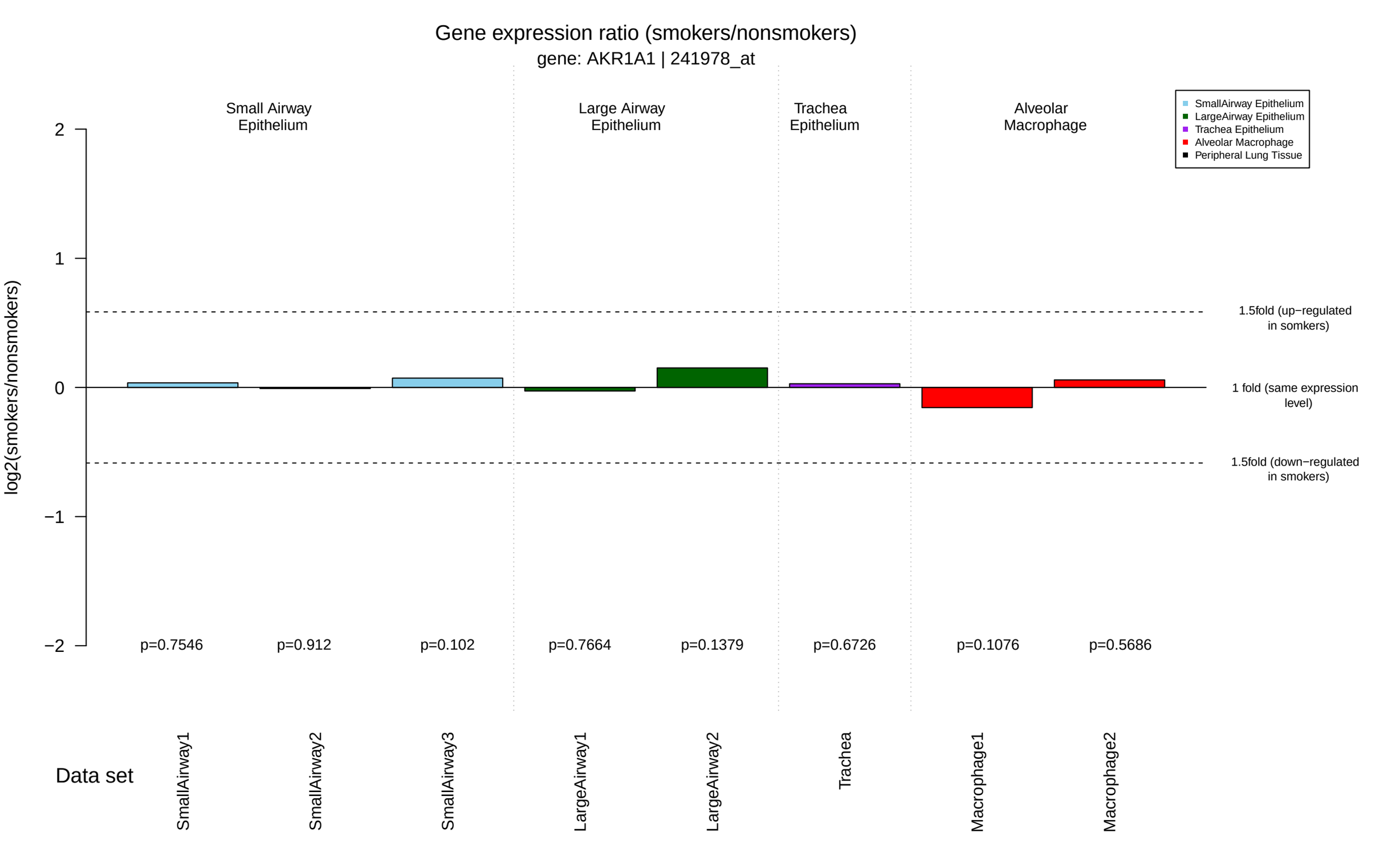
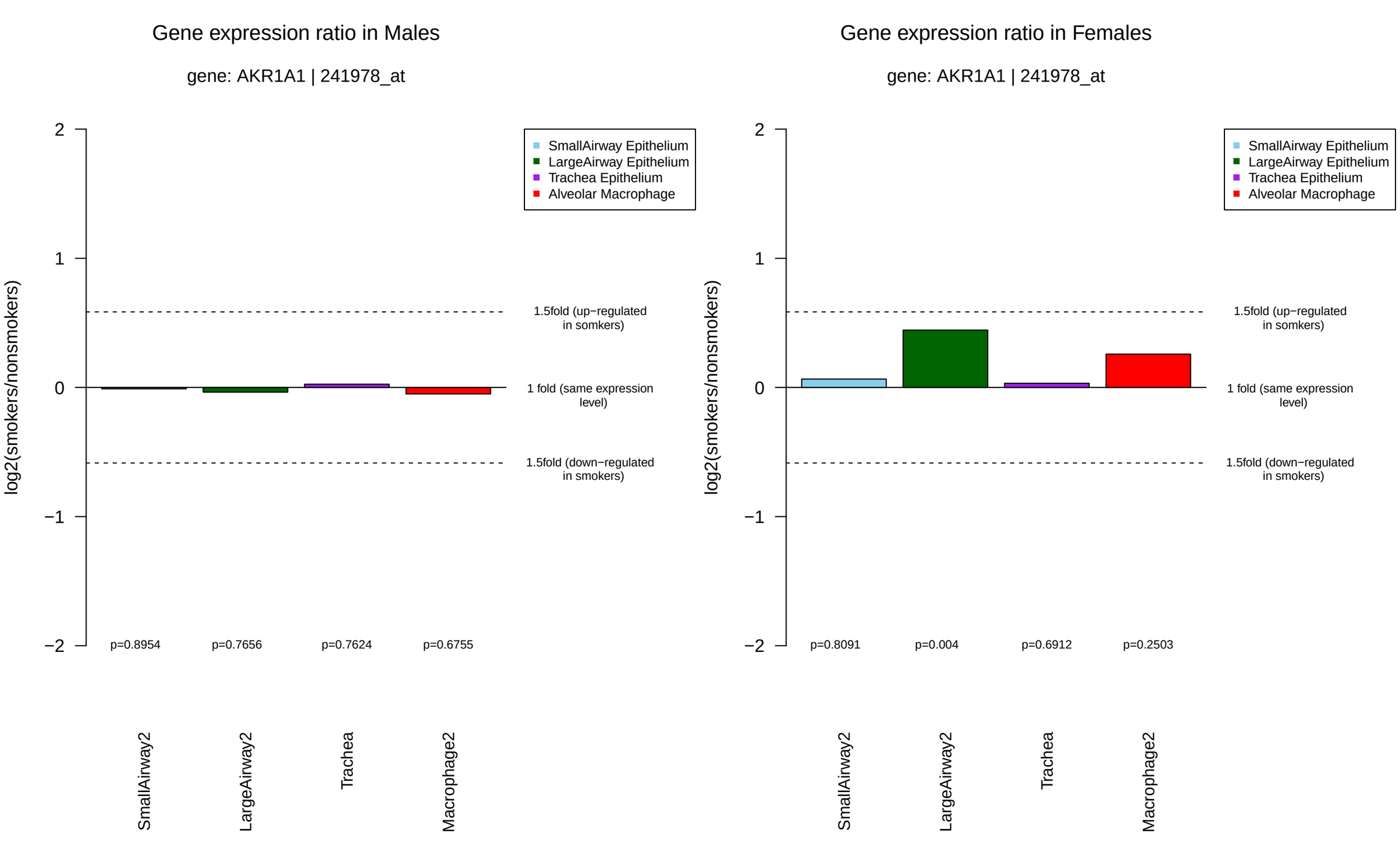
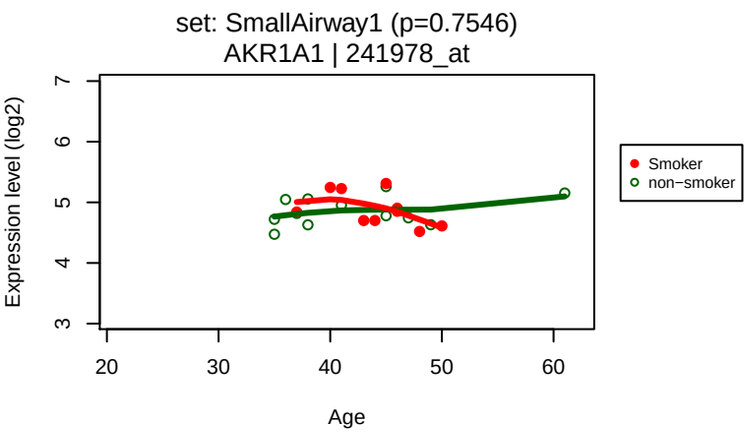
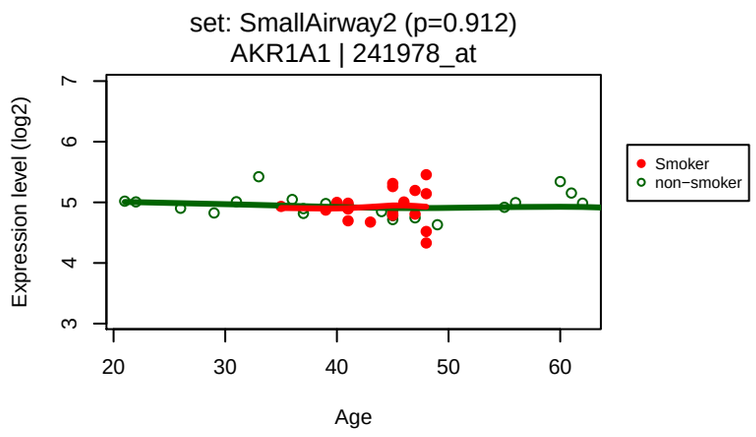
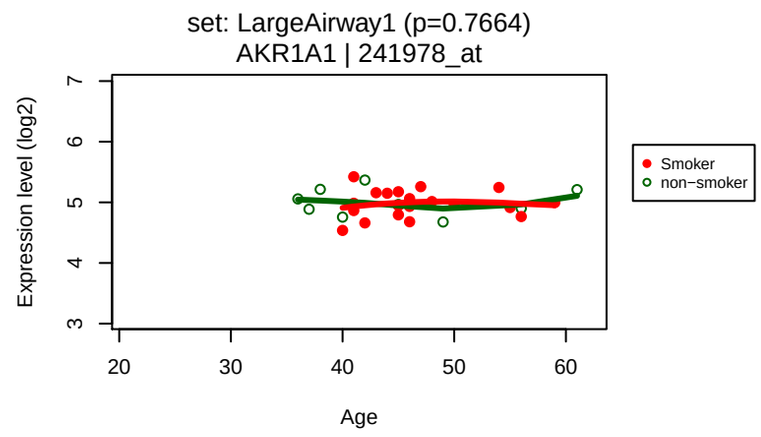

Gene expression of smokers and nonsmokers
gene: AKR1A1 | 241978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7546
p=0.912
p=0.102
p=0.7664
p=0.1379
p=0.6726
p=0.1076
p=0.5686
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

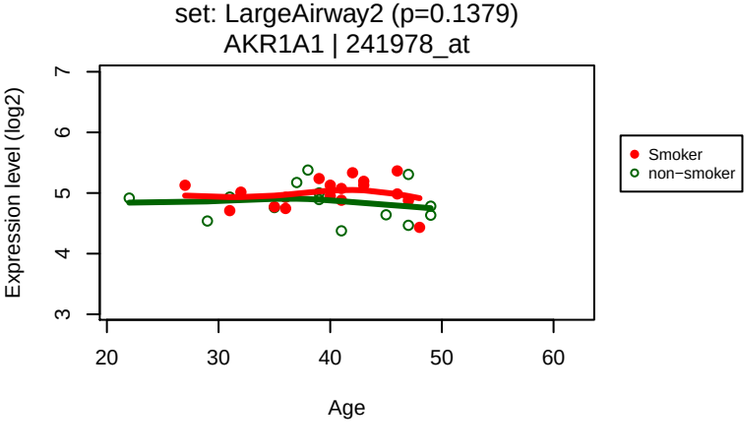

Gene expression ratio (smokers/nonsmokers)
gene: AKR1A1 | 241978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7546
p=0.912
p=0.102
p=0.7664
p=0.1379
p=0.6726
p=0.1076
p=0.5686
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

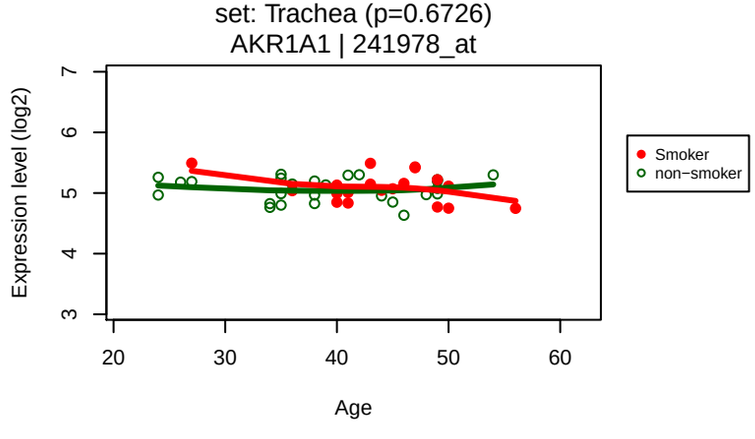

Gene expression ratio in Males
gene: AKR1A1 | 241978_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8954
p=0.7656
p=0.7624
p=0.6755
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR1A1 | 241978_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8091
p=0.004
p=0.6912
p=0.2503
SmallAirway2
LargeAirway2
Trachea
Macrophage2

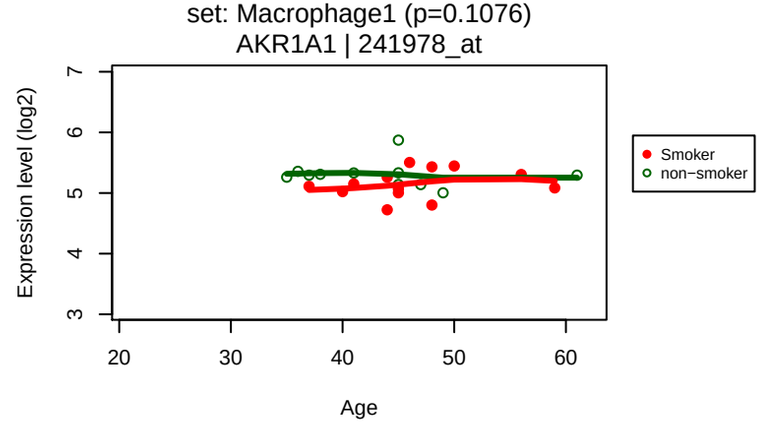

set: SmallAirway1 (p=0.7546)
AKR1A1 | 241978_at
Expression level (log2)
Age
Smoker
non−smoker

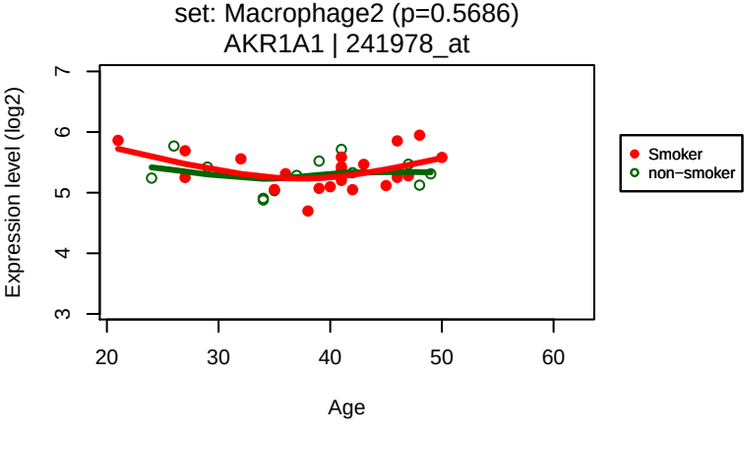

set: SmallAirway2 (p=0.912)
AKR1A1 | 241978_at
Expression level (log2)
Age
Smoker
non−smoker

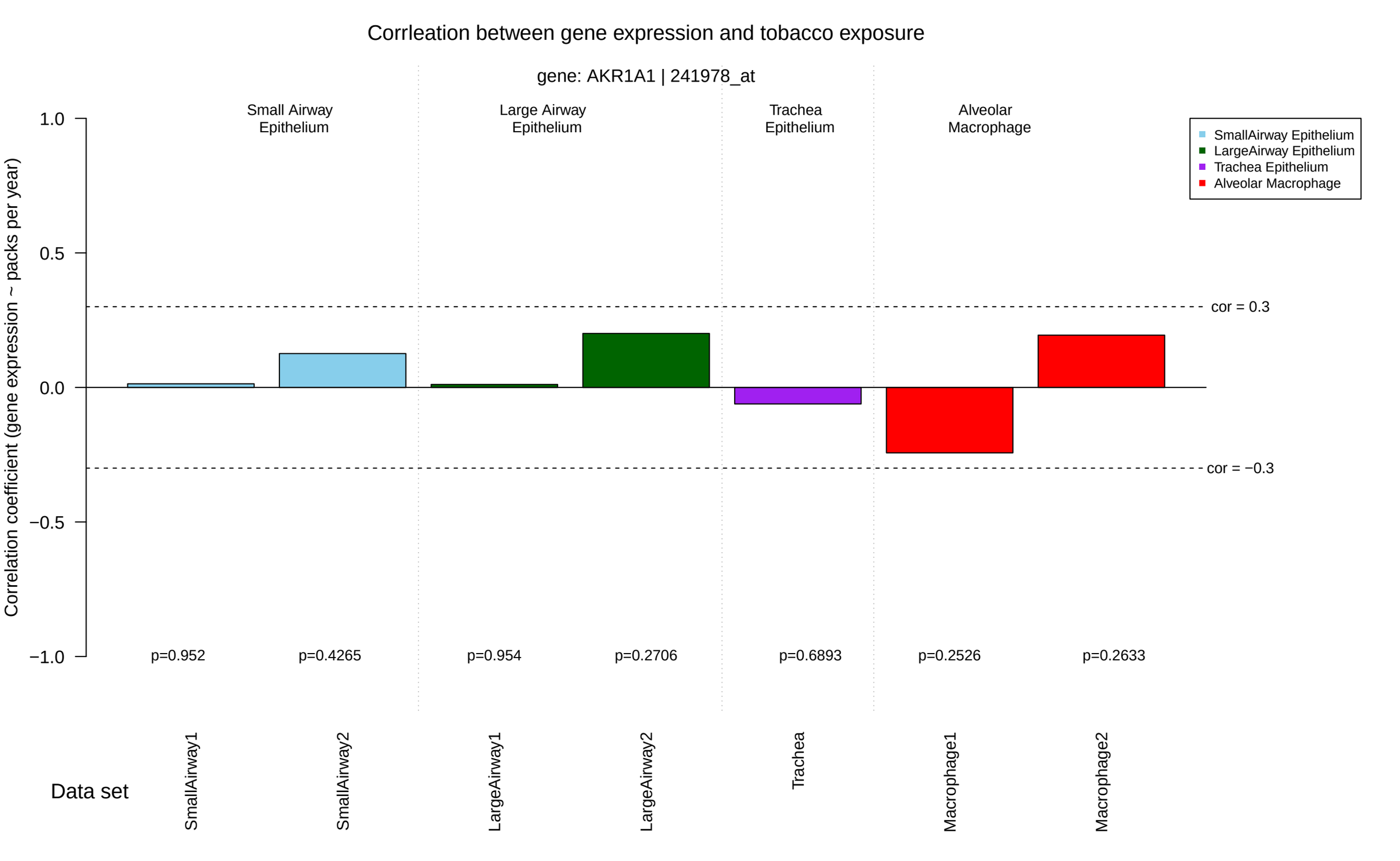

set: LargeAirway1 (p=0.7664)
AKR1A1 | 241978_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1379)
AKR1A1 | 241978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6726)
AKR1A1 | 241978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1076)
AKR1A1 | 241978_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5686)
AKR1A1 | 241978_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR1A1 | 241978_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.952
p=0.4265
p=0.954
p=0.2706
p=0.6893
p=0.2526
p=0.2633
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2